Bowen Yu

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Add: Yueyanglou District, Yueyang City, Hunan Province, China 414000
Research Interests: Large Language Models (LLMs), Algorithms, Cybersecurity, Data Mining, etc.

EDUCATION

The University of Manchester, UK **09/2026-09/2027**
- **Programme:** MSc Robotics
- **Core Modules:** Autonomous Mobile Robots, Reinforcement Learning, Cognitive Robotics and Computer Vision, Robotic Systems, etc.

Wenzhou-Kean University, China | GPA: 3.77/4.0 **09/2022-06/2026**
- **Programme:** BSc Computer Science (Major); Mathematics (Minor)
- **Language of Instruction:** English
- **Core Modules:** Data Structures & Algorithms; Computer Systems; Python Programming, etc.
- **Dissertation:** Transforming Malware Analysis with Explainable AI: A Detailed Review of Windows and Android Platforms

PUBLICATIONS

- Ullah, F., Mehmood, K., **Yu, B.W.**, et al. (2026). Advanced AI-Driven Privacy-Preserving Federated Learning with Generative Augmentation for Multimodal Emotion Recognition in 6G Networks. *Digital Communications and Networks*. **(Under Review)**
- Ullah, F., Mehmood, K., **Yu, B.W.**, et al. (2026). Trustworthy Privacy-Preserving Federated Learning with Deep Generative Augmentation for Multimodal Emotion Recognition in Healthcare. *IEEE Journal of Biomedical and Health Informatics*. **(Under Review)**
- Wu, T.H., Ullah, S., **Yu, B.W.**, Ullah, F., Wang, J.H., Gao, Z.Q., Yoshigoe, K. & Zhao, Y. (2026). A Survey on U-Net with Variants and Their Applications in Two-Dimensional Medical Image Segmentations. *Journal of Imaging*. 2026, 12(10), 497; https://doi.org/10.3390/jimaging12100497
- **Yu, B.W.**, Zheng, Q.T., Zhong, Z.Q., Yu, B., Zhao, Y. (2026). Role-Playing vs. Baseline Generation in Educational Dialogue: A Multi-Turn Comparative Study with Automated Scoring. *International Conference on Information Science And Big Data Analytics (ICISBDA-26)*. **(Accepted)**
- **Yu, B.W.**, et al. (2026). When Speech Meets Lips: Interpretable Audio-Visual Synchronization for L2 Pronunciation Assessment. arXiv. https://arxiv.org/abs/2609.06788.
- **Yu, B.W.**, Zhong, Z.Q., Liu, Y.S., Zhao, Y. (2026). Federated Transfer Learning for Personalized Household Financial Services. *The 2026 International Conference on Blockchain Technology and Foundation Models (BTFM)*, Shenzhen, China, 2026, pp. 61-66, doi: 10.1109/BTFM69084.2026.11484502.
- **Yu, B.W.**, Zhang, W.T., Liu, Y.S., Ullah, F., Yoshigoe, K., Ahmad, A., Obidallah, W.J., and Zhao, Y. (2026). Explainable AI for malware analysis: a systematic review of benchmark datasets, traffic-oriented detection, and interpretability methods. *PeerJ Computer Science* 12:e3902. https://doi.org/10.7717/peerj-cs.3902.
- Ehimwenma, K.E., Ji, H.W., Lin, J., Al Sharji, S.A., Cheraghy, M., **Yu, B.W.** (2024). Simulating Home Appliance Behavior and Operational Condition-Action Rules for Autonomous Intelligence. *2024 International Conference on Intelligent Computing and Next Generation Networks (ICNGN)*, Bangkok, Thailand, 2024, pp. 01-06, doi: 10.1109/ICNGN63705.2024.10871679.

HONOURS & AWARDS

HONOURS:
- Outstanding Graduate Award **06/2026**
- Dean's List Award (Six times) **2023-2026**

SCHOLARSHIPS:
- Zhejiang Provincial Government Scholarship **2025/26**
- The INM Inspiration and Innovation Scholarship **2025/26**
- Dean's Scholarship **2025/26, 2024/25**
- Research Innovation Scholarship **2024/25**
- Excellence in Culture, Sports & Arts Scholarship **2024/25, 2023/24**
- Excellence in Culture, Sports & Arts Team Scholarship **2023/24**

ACADEMIC AWARDS:

- Best Paper Award, 2024 ICNGN (International Conference on Intelligent Computing and Next Generation Networks) **11/2024**
- University Innovation Research Paper Award (Title: Artificial Intelligence Search on Self-Learning of Intelligent Systems: Explore Building Artificial Intelligence Systems with Self-Learning Capabilities) **01/2024**
- Excellent Professional Programming Award **12/2023**

RESEARCH EXPERIENCE

ByteDance AI Full-Stack Development Competition Project (Team Leader): 05/2026-06/2026
AI-Driven Competitive Analysis of Gaming Mice via Multi-Agent Systems

- **Objective:** Build a multi-agent system for automated competitive intelligence and benchmarking of AI-powered gaming mice.
- **Overview:** Designed a seven-node LangGraph workflow for research, evidence collection, analysis, validation, and reporting; developed a self-correcting QualityAgent feedback loop and a two-stage evidence verification framework; integrated MCP tools for real-time specification retrieval, pricing analysis, and competitor benchmarking.
- **Achievement:** Shortlisted twice in the ByteDance AI Full-Stack Development Competition, with the project selected for the final presentation and recognised as an **Outstanding Project Team**.

China-U.S. Young Maker Competition (Finalist) Project (Team Member): 04/2025-05/2025
AI-Powered Smart Flow Control Device for Traditional Chinese Medicine

- **Objective:** Develop an AI-enabled smart flow control device for personalised Traditional Chinese Medicine (TCM) delivery using real-time health monitoring.
- **Overview:** Co-developed a hardware-software system integrating wearable biometric data, AI-driven health analysis, and automated flow control to personalise herbal formulations for home and office use.
- **Achievement:** Reached the finals of the China-U.S. Young Maker Competition, demonstrating the feasibility of AI-assisted personalised TCM delivery and intelligent health management.

BSc Dissertation: Transforming Malware Analysis with Explainable AI: A Detailed Review of Windows and Android Platforms | Supervisor: Dr. Yue Zhao **2024-2026**

- **Objective:** Review recent advances in malware analysis for Windows and Android, with a focus on explainable AI (XAI) methods for improving detection transparency and reliability.
- **Methods:** Conducted a systematic literature review of benchmark datasets, deep learning approaches, and XAI techniques; analysed the application of LIME and SHAP for malware detection and interpretability.
- **Key Results:** Identified the strengths and limitations of XAI-assisted malware detection and highlighted key challenges and future directions for scalable and trustworthy malware analysis systems.

Simulating Home Appliance Behavior and Operational Condition-Action Rules for Autonomous Intelligence (Published and Awarded Best Paper at ICNGN 2024) 03/2023-04/2023

- **Objective:** Develop an autonomous IoT framework that enables home appliances to perceive environmental conditions and execute condition-action rules independently.
- **Methods:** Modelled and simulated autonomous appliance behaviours, implementing modular object-oriented programs for sensor and appliance agents to support perception, communication, and autonomous decision-making.
- **Outcome:** Demonstrated that sensor-driven appliance agents can autonomously execute condition-based actions, providing a practical framework for intelligent and self-managing IoT home systems.

WORK EXPERIENCE

RedNote (Xiaohongshu), Shanghai, China 06/2026-09/2026
Product Engineer | AI Application Full-Stack Development, Lingxi AI

- Designed and implemented an automated AI agent quality reporting system, integrating data collection, KPI analysis, case identification, and team-wide notification workflows.
- Built fully automated scheduled pipelines leveraging Lingxi AI platform APIs to monitor daily PV/UV, dialogue trends, and user retention metrics.
- Developed observability dashboards, anomaly alerting, and fault-tolerant mechanisms to ensure reporting accuracy, operational transparency, and high system reliability.
- Designed and deployed enterprise-grade AI agent skill packages (e.g., for Xiaomi 3D digital products and Louis Vuitton), enabling autonomous data extraction and automated front-end report generation.

Architecture Matrix Technology Co., Ltd., Guangzhou, China 04/2026-06/2026
AI Product Manager | AI Agent Development, GEO & Short-Video Generation Team

- Contributed to the end-to-end development of a commercial geospatial SaaS platform, from system architecture design to pre-production deployment.

- Led the 0-to-1 development of an AI-driven multi-agent short-video content generation system for million-follower creators, coordinating scripting, evaluation, storyboard design, and production workflows.
- Built a multi-vendor LLM evaluation framework (Qwen, Doubao, DeepSeek) with automated health monitoring, fault isolation, and model performance backtesting.
- Developed scalable backend services using FastAPI, Redis, PostgreSQL, and Docker, enabling robust task scheduling, real-time analytics, and secure end-to-end data pipelines.

Shining 3D Tech Co., Ltd., Hangzhou, China **06/2024-09/2024**
Algorithm Engineer | Algorithms Department

- Contributed to the development, optimisation, and maintenance of active and passive 3D reconstruction algorithms, improving accuracy and computational efficiency across diverse scenarios.
- Designed, tested, and optimized AI models for computer vision-based inspection tasks.
- Participated in technical reviews and cross-functional planning, providing algorithmic feasibility analysis and product insights.
- Built Python-based data processing pipelines and SQL workflows to manage and analyze large-scale 3D datasets, supporting data-driven decision-making.

China Petroleum & Chemical Corporation (Sinopec), Hunan, China **12/2023-02/2024**
Cybersecurity Engineer | IT Technical Services Department

- Hardened and secured two Windows Server environments by disabling unused ports, implementing role-based access controls, and optimizing firewall policies.
- Contributed to SQL Server security policy development, successfully preventing and logging five unauthorized access attempts.
- Detected and mitigated two anomalous traffic nodes using Wireshark and firewall log analysis, improving network stability and incident response capability.
- Reviewed and processed 150 VPN and system access requests; streamlined approval workflows, reducing processing time by 30%.

Yueyang Museum, Hunan, China **06/2023-08/2023**
Backend Development Engineer | IT Department

- Co-developed and deployed an exhibit information management module for the museum's backend system, supporting structured data management for 200+ artifacts.
- Optimised system performance using object-oriented design and multithreading in Java, reducing average response time by 20%.
- Resolved three critical functional defects during iterative releases and contributed to technical documentation, improving code readability and long-term maintainability.

SKILLS

- **Languages:** English (proficient | IELTS: 6.5); Chinese (native)
- **Programming:** Java, Python, R, SQL
- **Web Technologies:** HTML, CSS
- **ML Frameworks:** TensorFlow, PyTorch
- **Others:** National Second-level Volleyball Athlete